

Eigen Vectors & Eigen Values

What is a matrix

A matrix is a structured way to store numbers that represent a transformation.

→ A matrix tells how inputs change into outputs

Matrix as a function

In linear algebra: A matrix represents a function that transforms vectors.

Input vector $\rightarrow$ Matrix $\rightarrow$ Output vector

A vector is a point, a direction, a feature set

$$x = [2, 1]$$

Linear Transformation:-

A linear Transformation changes vector while:

- Keeping straight line straight
- Preserving the origin (0 stays 0)

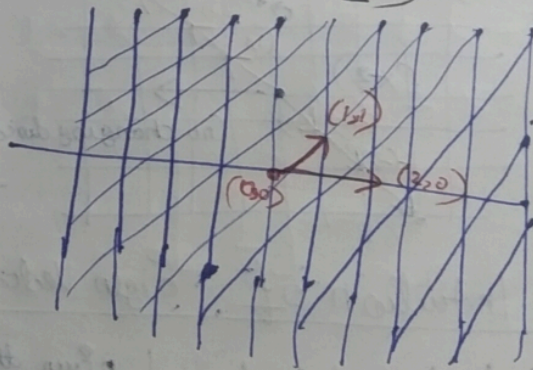
Ex:- Linear transformations

- Rotation
- Stretching
- Projection (PCA)
- Scaling
- Shearing

how matrices perform Linear Transformation

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x = [1, 1] \Rightarrow Ax = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- x-direction scaled by 2
- y-direction - unchanged



What is an Eigenvector:-

• An Eigenvector is a Special direction that does not change direction when a matrix (transformation) is applied to it.

→ only its length (magnitude) may change.

Visualization :- Geo Gebora

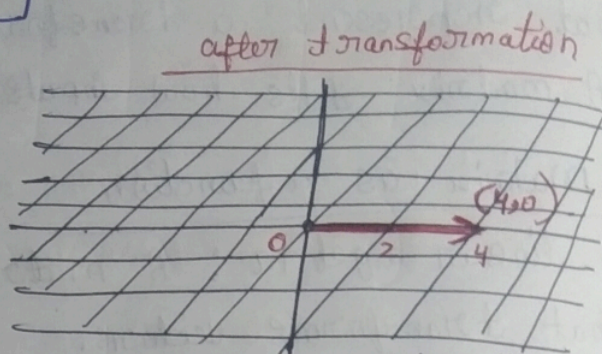
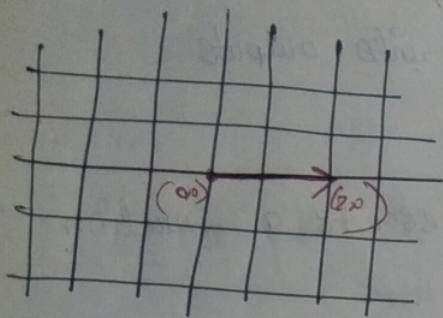
$$a_{11} = \boxed{2}$$

$$a_{12} = \boxed{1}$$

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$a_{21} = \boxed{0}$$

$$a_{22} = \boxed{1}$$



- i) Vector in same direction
- ii) but magnitude changed to (4,0)

Eigen Value :-

An Eigen Value tells how much an eigen vector is stretched or squashed by a matrix.

→ If Eigen vector = direction

→ Then Eigen value = Strength along that direction

$$A = \lambda \vec{v}$$

$\vec{v} \Rightarrow$ Eigen vector [direction stays same]

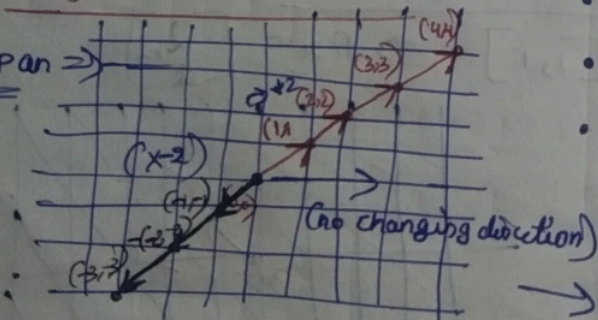
$\lambda \Rightarrow$ Scaling factor (Eigen value) (magnitude)

$$\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$\lambda = 2$

Eigen Vectors or Span

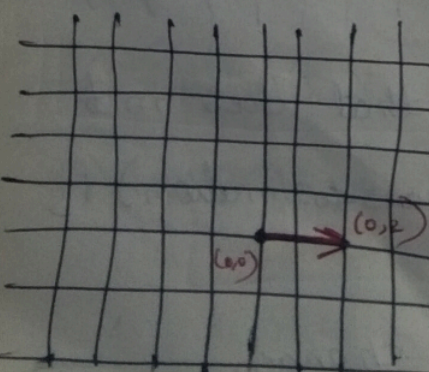
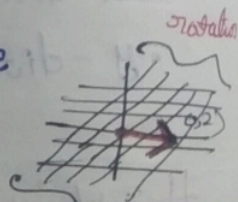
Span $\Rightarrow$



- multiplying vectors by any number
- add them
- all such results belongs to Span

→ Span is only a line

Intuition of Eigen Vectors & Eigen Values



• Even though you perform any linear transformation the axis will change but (direction) Eigen vector remains same

after transformation



when transform vector not change only the axis is rotated

How to calculate Eigen vector & Eigen Values:-

matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3}$, matrix $B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}_{3 \times 2}$

$(2 \times 2) A \times B = \begin{bmatrix} 7+18+36 & 8+20+36 \\ 28+45+66 & 32+50+72 \end{bmatrix} \Rightarrow \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$

$$A\vec{v} = \lambda\vec{v}$$

$A \rightarrow$ matrix, $I =$ Identity matrix
 $\vec{v} \rightarrow$ vector
 $\lambda \rightarrow$ scalar

$$A\vec{v} - \lambda\vec{v} = 0$$

$$(A - \lambda I)\vec{v} = 0$$

Why determinant = 0

$$(A - \lambda I)\vec{v} = 0$$

Non-trivial

// for non-zero vector $\vec{v}$ to exist

i) Trivial Solution:- $\vec{v} = 0$ // not allowed for Eigen vector
 // Eigen vector must be non-zero

ii) Non Trivial Solution:- $\vec{v} \neq 0$

• has non zero solutions, iff matrix A is Singular

• A matrix is singular if $\det(M) = 0$

So, $\det(A - \lambda I) = 0$

Why determinant $\neq 0$ fails

$$\text{if } \det(M) \neq 0$$

Then • matrix is invertible

• multiply both by M^{-1} ; $\vec{v} = 0$

$\vec{x}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow \det(N) = 1 - 0$$

$$\det(N) = 1, \neq 0$$

$$N\vec{v} = 0, \vec{v} = 0, \text{ No Eigen vector}$$

• Trivial solution $\Rightarrow$ no Eigen vector

$$\det(A - \lambda I) = 0$$

because we want non-zero

solutions to $(A - \lambda I)\vec{v} = 0$, and a homogeneous

system has non trivial solutions only when the determinant is zero

$$\boxed{\det(A - \lambda I) = 0} \rightarrow \text{This Equation gives Eigen values}$$

Ex: $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, $\Rightarrow A - \lambda I \Rightarrow \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \Rightarrow \begin{bmatrix} 4-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix}$

Step 2: Take determinant

$$\det(A - \lambda I) \Rightarrow (4-\lambda)(3-\lambda) - (2 \times 1)$$

$$\Rightarrow 12 - 4\lambda - 3\lambda + \lambda^2 - 2$$

$$\Rightarrow \boxed{\lambda^2 - 7\lambda + 10} \text{ \{ take factors \}}$$

$$\Rightarrow \lambda^2 - 5\lambda - 2\lambda + 10$$

$$\Rightarrow (\lambda - 5)(\lambda - 2)$$

Eigen values $\Rightarrow \lambda_1 = 5, \lambda_2 = 2$

find Eigen Vectors :-

i) for $\lambda = 5 : (A - 5I)V = 0$

$$\left(\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{array}{l|l} -x + 2y = 0 & x + 2y = 0 \\ x = 2y & x = 2y \end{array}$$

$$V_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

ii) $\lambda = 2 : (A - 2I)V$

$$\left(\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{array}{l|l} 2x + 2y = 0 & 1x + 1y = 0 \\ \boxed{x = -y} & \boxed{x = -y} \end{array}$$

$$V_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

lies on same line
Same direction

Ex: $(1, -1)$
 $(2, -2)$
 $(-3, 3)$

Properties of Eigen Vectors & Eigen Values :-

i) Eigen Vectors are non-zero Vectors

$$\boxed{AV = \lambda V, V \neq 0}$$

• Zero vector is never an eigenvector

• Because direction must

→ Eigen Vectors must be non-zero
they represent the directions

ii) Eigen Vectors can be +ve, -ve or zero:

- Positive → stretch
- Negative → flip + stretch
- Zero → collapse into lower dimension

→ In PCN:- Eigenvalues ≥ 0 (Covariance matrix)

n - features $\Rightarrow$ n - Eigen values

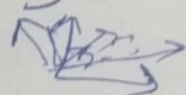
iii) Eigen vectors are linearly independent:

- This allows diagonalization
- Dimensionality reduction

iv) Orthogonality:

→ for real symmetric matrices (like Covariance matrix)

- Eigenvectors are orthogonal
- This is why PCA axes are perpendicular



v) Zero EigenValue Meaning:

- if $\lambda = 0$: Then
- Transformation Squashes data
 - feature is redundant
 - dimension can be removed

PCA:- removes such directions.

Eigen Vectors in PCA:-

$\begin{matrix} \text{cgpa} / \text{iq} & | & \text{lpa} \\ \hline - & - & - \\ - & - & - \end{matrix} \Rightarrow$

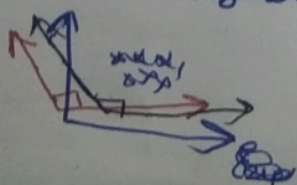
Performing feature Extraction

calculate Covariance matrix

features = $n \times n$
(2×2)

how it is varying
 $\begin{bmatrix} \text{Var}(\text{iq}) & \text{Cov}(\text{iq}, \text{cgpa}) \\ \text{Cov}(\text{cgpa}, \text{iq}) & \text{Var}(\text{cgpa}) \end{bmatrix}$
Explains correlation b/w iq & cgpa
how the spread of Var

Orthogonal



Calculate Eigen Vectors

these are new features extracted

Max Variance Capture